
Beyond SIFT: An End-to-End Pipeline for Robust Feature Matching in View-Dependent Acoustic Vision

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Abstract

Underwater robotic perception using side-scan sonar (SSS) is severely limited by viewpoint-dependent backscatter, acoustic shadows, and speckle noise, rendering traditional feature detectors such as SIFT unreliable for robust cross-view matching. This work presents an end-to-end pipeline for viewpoint-invariant feature matching in SSS imagery. We apply DINOv3 self-distillation to pre-train a ConvNeXt-tiny backbone on ~ 1 million unlabeled tiles from the BenthicCat dataset, enabling domain-specific, self-supervised representation learning tailored to acoustic data. The resulting dense, semantically consistent features support a robust matching pipeline that combines multi-scale hypercolumn descriptors, cosine similarity, sub-pixel refinement, and RANSAC-based geometric verification. Preliminary qualitative results demonstrate successful clustering of seafloor textures despite heavy noise and radiometric augmentations. The approach offers a lightweight, edge-deployable solution for autonomous underwater navigation and large-scale seafloor mapping.

Keywords: side-scan sonar, self-supervised learning, feature matching, ConvNeXt, DINOv3, underwater robotics

1 Introduction

The underwater robotic perception is severely limited by the physical properties of the marine environment. Electromagnetic radiation attenuates rapidly, making optical sensors effective only at very short ranges (0–10 m) Lei et al. [2025b], Katrusha et al. [2025]. As a result, side-scan sonar (SSS) has become the dominant modality for large-scale seafloor mapping, target detection, and autonomous navigation Burguera and Oliver [2016], Rajani et al. [2025]. SSS systems produce high-resolution 2D “waterfall” images from acoustic backscatter intensity, revealing detailed topographic and textural information Lei et al. [2025b], Burguera and Oliver [2016].

Despite mature acoustic hardware, real-time autonomous interpretation via machine learning remains a key challenge Rajani et al. [2025], Lei et al. [2025b]. Traditional analytical methods struggle with the strong viewpoint dependency, signal degradation, and speckle noise inherent in acoustic imaging Lei et al. [2025b,a]. Recent advances in deep self-supervised learning (SSL) enable robust feature extraction from large-scale unlabeled sonar data, reducing dependence on expensive manual annotations Siméoni et al. [2025], Preciado-Grijalva et al. [2022].

1.1 Challenges in SSS Perception

SSS image formation differs fundamentally from optical perspective geometry. Backscatter intensity $I(x, y)$ is modeled as

$$I(x, y) = \rho(x, y) \cdot \cos(\theta(x, y)) \cdot L(x, y), \quad (1)$$

where ρ is intrinsic seabed reflectivity, θ is the local incidence angle, and L accounts for propagation loss Lei et al. [2025a], Burguera and Oliver [2016].

This model produces strongly viewpoint-dependent images: different survey directions yield different intensity patterns and inverted shadows. Acoustic shadows arise from occlusion by topographic features, while coherent speckle noise introduces granular artifacts that degrade fine detail Katrusha et al. [2025], Preciado-Grijalva et al. [2022]. These characteristics cause poor repeatability of traditional keypoints (SIFT, SURF) and high matching error in underwater settings Lei et al. [2025b], Katrusha et al. [2025], Fu et al. [2025].

1.2 Contributions

This work addresses core challenges in underwater acoustic perception through modern foundation-model techniques and efficient convolutional architectures. The main contributions are:

- Application of DINOv3 self-distillation to the *BenthiCat* dataset ($\sim 1\text{M}$ tiles) for domain-specific pre-training on unlabeled sonar data, closing the cross-modal degradation gap from natural images to acoustics Rajani et al. [2025], Siméoni et al. [2025].
- Adoption of ConvNeXt-tiny as a lightweight, modernized convolutional backbone that incorporates Transformer-inspired design elements (large kernels, inverted bottlenecks, LayerNorm) while preserving ConvNet efficiency and enabling edge deployment on AUVs Liu et al. [2022], Siméoni et al. [2025].
- A robust end-to-end feature matching pipeline based on dense self-supervised features from the ConvNeXt backbone for viewpoint-invariant, geometrically consistent correspondences even in low-texture and shadowed regions.

1.3 State of the Art in Feature Extraction and Matching

Reliable cross-view correspondence in SSS remains difficult due to geometric distortion and speckle noise Lei et al. [2025b], Katrusha et al. [2025]. Handcrafted descriptors (SIFT, SURF, AKAZE) perform poorly because of their sensitivity to local intensity gradients disrupted by speckle and view-dependent backscatter Lowe [2004], Lei et al. [2025b], Fu et al. [2025].

Current best methods use learned detectors (e.g., SuperPoint) and Transformer-based matchers (LightGlue, LoFTR) Katrusha et al. [2025], Fu et al. [2025]. LightGlue performs efficient sparse matching via iterative self- and cross-attention with adaptive computation. LoFTR offers dense, detector-free matching with high coverage in texture-poor areas, at higher computational cost Katrusha et al. [2025], Fu et al. [2025].

For cross-modal alignment (e.g., SSS with MBES), symmetric epipolar distance is commonly used to enforce geometric consistency:

$$d(p_1, p_2) = \frac{(p_2^\top F p_1)^2}{\|F p_1\|^2} + \frac{(p_1^\top F^\top p_2)^2}{\|F^\top p_2\|^2}, \quad (2)$$

where F is the fundamental matrix Fu et al. [2025].

2 Data Modality and the BenthiCat Dataset

2.1 Side-Scan Sonar Principles

Side-scan sonar is the primary modality for large-scale underwater perception, as acoustic waves propagate far better than electromagnetic radiation in seawater Lei et al. [2025b], Katrusha et al. [2025]. SSS systems use laterally mounted transducers on a moving platform (e.g., AUV) to emit fan-shaped pulses perpendicular to the trajectory. Returned backscatter intensity is recorded as a function of time-of-flight and converted to range assuming a sound speed of 1500 m/s, producing 2D “waterfall” images of seafloor acoustic reflectivity Burguera and Oliver [2016], Rajani et al. [2025].

Image formation follows a Lambertian model:

$$I(x, y) = \rho(x, y) \cdot \cos(\theta(x, y)) \cdot L(x, y), \quad (3)$$

where ρ is intrinsic seabed reflectivity, θ is the local incidence angle, and L is path loss due to spreading and absorption Lei et al. [2025a], Burguera and Oliver [2016], Lei et al. [2025b].

This results in strongly viewpoint-dependent imagery: different survey directions produce distinct intensity patterns and shadow geometries. Acoustic shadows arise from occlusion by topography or objects, while speckle noise—multiplicative granular interference from coherent wave scattering—further degrades detail Katrusha et al. [2025], Preciado-Grijalva et al. [2022]. These challenges require robust, view-invariant feature extractors such as modernized convolutional networks Liu et al. [2022], Siméoni et al. [2025].

2.2 The BenthicCat Dataset

The *BenthicCat* dataset supports benthic habitat mapping with co-registered opti-acoustic data collected along the Catalan coast, covering $> 1900 \text{ km}^2$ Rajani et al. [2025]. Acoustic data were acquired using a dual-frequency Klein 3000H SSS at 500 kHz, achieving $\sim 8.5 \text{ cm}$ average slant-range resolution Rajani et al. [2025], Lei et al. [2025a].

It contains ~ 1 million unannotated SSS image tiles suitable for self-supervised pre-training with DINOv3 Rajani et al. [2025], Siméoni et al. [2025]. For supervised evaluation and fine-tuning, $\sim 36,000$ tiles are annotated with pixel-wise segmentation masks across 12 habitat classes (e.g., *Posidonia oceanica*, sand ripples, rock outcrops) Rajani et al. [2025].

Additionally, 178,000 co-registered optical images were captured by the Girona1000 AUV at low altitude (3.5–4.5 m), georeferenced to corresponding SSS tiles. This enables cross-modal learning and verification of acoustic features against optical ground truth Rajani et al. [2025].

2.3 Data Preprocessing

Raw 12-bit SSS waterfall data undergo the following preprocessing steps:

- Logarithmic compression and normalization to $[0, 1]$ to stabilize training and preserve low-intensity detail:

$$I'_{\text{normalized}} = \frac{\ln(1 + I_{\text{raw}})}{\max(\ln(1 + I_{\text{raw}}))}. \quad (4)$$

- Slant-range correction under the flat-floor assumption to convert slant range r_s to ground range r_g :

$$r_g = \sqrt{r_s^2 - h^2}, \quad (5)$$

removing nadir compression and blind zone Burguera and Oliver [2016], Rajani et al. [2025].

- Tiling into overlapping 384×384 pixel patches with 192-pixel stride in both along- and across-track directions to preserve structural continuity and leverage ConvNeXt translation equivariance Liu et al. [2022], Rajani et al. [2025].

3 Backbone Architecture: ConvNeXt

For autonomous underwater perception with Side-Scan Sonar, the backbone must handle severe geometric distortions and non-linear intensity variations Lei et al. [2025b], Katrusha et al. [2025]. While Vision Transformers offer strong performance, their quadratic complexity hinders deployment on resource-constrained AUVs Liu et al. [2022], Siméoni et al. [2025]. We therefore adopt **ConvNeXt-tiny** as the feature extractor within the **DINOv3** self-supervised framework Liu et al. [2022], Siméoni et al. [2025].

ConvNeXt modernizes the ResNet design space by retaining convolutional inductive biases (translation equivariance, local connectivity) while incorporating successful elements from hierarchical Transformers Liu et al. [2022].

3.1 Modernizing ResNet

The ConvNeXt design begins with an improved training recipe (AdamW optimizer, 300 epochs, Mixup, CutMix, RandAugment), raising baseline ResNet-50 ImageNet top-1 accuracy from 76.1%

to 78.8% Liu et al. [2022]. Structural changes then follow five stages: macro design, ResNeXt-style cardinality, inverted bottlenecks, large kernels, and micro refinements.

The convolutional locality of ConvNeXt aligns well with the ping-by-ping, locally correlated nature of SSS data, making it suitable for sediment texture and object boundary detection Burguera and Oliver [2016], Lei et al. [2025a].

3.2 Key Design Choices

Macro design ConvNeXt-tiny uses the Swin-T stage compute ratio (3, 3, 9, 3 blocks) and replaces the ResNet 7×7 stride-2 stem + max-pool with a 4×4 stride-4 patchify stem, mirroring ViT patch embedding Liu et al. [2022].

Depthwise conv + inverted bottleneck Depthwise 7×7 convolutions (moved before 1×1 layers) separate spatial and channel mixing, similar to self-attention. An inverted bottleneck expands the MLP hidden dimension to $C_{\text{hidden}} = 4 \cdot C_{\text{input}}$, placing expensive operations on fewer channels Liu et al. [2022].

Large kernels The 7×7 depthwise kernel significantly enlarges the receptive field, approaching Swin Transformer window sizes and capturing large-scale topographic features and acoustic shadows in SSS imagery Liu et al. [2022], Lei et al. [2025b].

Micro refinements - Replace ReLU with GELU - Use fewer activations and norms per block (one GELU, one LayerNorm) - Replace BatchNorm with LayerNorm for training stability Liu et al. [2022]

3.3 Computational Efficiency

ConvNeXt-tiny achieves 82.1% ImageNet top-1 accuracy at 4.5 GFLOPs, outperforming Swin-T (81.3%) at similar compute Liu et al. [2022]. It offers up to 49% higher inference throughput than Swin-T on A100 GPUs (TF32, channels-last) Liu et al. [2022].

In the DINOv3 framework, ConvNeXt-tiny enables efficient high-resolution feature extraction on edge devices, supports low-bitwidth quantization better than Transformers, and produces crisp, semantically consistent dense features across varying sonar resolutions Siméoni et al. [2025], Rajani et al. [2025], Fu et al. [2025].

4 Self-Supervised Learning with DINOv3

Visual representation learning has shifted from supervised to self-supervised methods that exploit unlabelled data Khan et al. [2025]. This is particularly important for SSS imaging, where data acquisition and expert annotation are expensive and time-consuming Rajani et al. [2025], Preciado-Grijalva et al. [2022].

DINOv3 employs a self-distillation framework combining global discriminative alignment, local masked reconstruction, feature regularization, and Gram anchoring to learn robust representations for SSS data.

4.1 Self-Distillation Framework

DINOv3 uses a teacher-student architecture where the teacher is an exponential moving average (EMA) of the student weights Siméoni et al. [2025], Khan et al. [2025]. The student matches the teacher’s output distributions instead of using contrastive pairs.

The pre-training loss is:

$$\mathcal{L}_{\text{Pre}} = \mathcal{L}_{\text{DINO}} + \mathcal{L}_{\text{iBOT}} + \mathcal{L}_{\text{Gram}} + 0.1 \cdot \mathcal{L}_{\text{DKoLeo}} \quad (6)$$

A final layer normalization is applied to backbone outputs of both global and local crops to stabilize dense feature learning late in training Siméoni et al. [2025].

4.1.1 Global Discriminative Loss ($\mathcal{L}_{\text{DINO}}$)

The DINO objective enforces view-invariance at the image level using a multi-crop strategy: the student sees two global crops and several local crops, while the teacher sees only global crops Oquab et al. [2024], Siméoni et al. [2025].

Output distributions are obtained via a projection head + softmax over K dimensions. To avoid collapse, DINOv3 uses Sinkhorn-Knopp centering instead of the original centering-sharpening Caron et al. [2021], Oquab et al. [2024]. The loss aligns the student distribution P_s with the centered teacher distribution P_t :

$$\mathcal{L}_{\text{DINO}} = - \sum_{v \in V} \sum_{v' \in V_{\text{global}}} P_t(x_{v'}) \log P_s(x_v) \quad (7)$$

This promotes features invariant to geometric distortions and speckle noise typical in SSS imagery Rajani et al. [2025].

4.1.2 Masked Latent Reconstruction ($\mathcal{L}_{\text{iBOT}}$)

The iBOT objective adds dense prediction capability by reconstructing masked patch tokens using the teacher’s unmasked representations Oquab et al. [2024], Siméoni et al. [2025]:

$$\mathcal{L}_{\text{iBOT}} = - \sum_{i \in \mathcal{M}} P_{t_i}^{\text{patch}} \log P_{s_i}^{\text{patch}} \quad (8)$$

Sinkhorn-Knopp centering is also applied to patch projections. This helps the model infer missing sonar information from acoustic shadows and seabed textures, improving local feature quality on datasets such as BenthicCat Rajani et al. [2025], Siméoni et al. [2025].

4.1.3 Feature Regularization ($\mathcal{L}_{\text{DKoLeo}}$)

KoLeo regularization prevents feature collapse by encouraging uniform hyperspherical distribution within each batch Oquab et al. [2024], Siméoni et al. [2025]:

$$\mathcal{L}_{\text{KoLeo}} = - \frac{1}{n} \sum_{i=1}^n \ln \left(\min_{j \neq i} \|z_i - z_j\| \right) \quad (9)$$

This maximizes minimum distances between samples, improving class separability in downstream benthic tasks.

4.2 Gram Anchoring

Long training schedules and high resolutions often degrade dense feature maps, causing patch-level inconsistency and noisy representations that harm SSS matching and registration Siméoni et al. [2025], Katrusha et al. [2025].

DINOv3 introduces Gram anchoring to preserve local structure Siméoni et al. [2025]. The Gram matrix captures pairwise correlations of L_2 -normalized patch features. The student is regularized by matching its Gram matrix to that of a stable “Gram teacher” (a checkpoint from earlier training):

$$\mathcal{L}_{\text{Gram}} = \|X_S X_S^\top - X_G X_G^\top\|_F^2 \quad (10)$$

In the refinement phase the full loss becomes:

$$\mathcal{L}_{\text{Ref}} = \mathcal{L}_{\text{DINO}} + \mathcal{L}_{\text{iBOT}} + \mathcal{L}_{\text{Gram}} + 0.1 \cdot \mathcal{L}_{\text{DKoLeo}} \quad (11)$$

Gram anchoring stabilizes dense features, accelerates iBOT convergence, and significantly improves performance on dense SSS tasks such as feature matching and segmentation while preserving fine sediment textures Siméoni et al. [2025], Rajani et al. [2025].

5 SSS Feature Extraction and Matching Pipeline

Accurate correspondence between heterogeneous side-scan sonar (SSS) images is essential for large-scale seafloor mapping and autonomous underwater navigation Lei et al. [2025b], Katrusha et al. [2025]. Unlike optical images, SSS data consists of range-based intensity profiles affected by seabed reflectivity, grazing angle, acoustic shadowing, and speckle noise, making traditional gradient-based descriptors (SIFT, SURF) poorly repeatable across overlapping swaths Burguera and Oliver [2016], Katrusha et al. [2025], Lei et al. [2025b].

We propose a pipeline that uses self-supervised dense feature extraction followed by robust outlier rejection to overcome radiometric instability and severe viewpoint-dependent distortions.

5.1 Dense Feature Extraction and Matching

We employ ConvNeXt-tiny Liu et al. [2022] as the backbone to extract multi-scale dense features, capturing both fine textures and broader semantic structures. Features are taken from stages $s \in \{2, 3, 4\}$ of the network, denoted $\mathbf{F}^{(s)} \in \mathbb{R}^{C_s \times H_s \times W_s}$.

The feature maps from stages 3 and 4 are upsampled to the resolution of stage 2 using bilinear interpolation:

$$\tilde{\mathbf{F}}^{(s)} = \begin{cases} \mathbf{F}^{(s)} & s = 2 \\ \mathcal{U}(\mathbf{F}^{(s)}, H_2, W_2) & s \in \{3, 4\} \end{cases} \quad (12)$$

Each upsampled map is L_2 -normalized per spatial location:

$$\hat{\mathbf{f}}_{i,j}^{(s)} = \frac{\mathbf{f}_{i,j}^{(s)}}{\|\mathbf{f}_{i,j}^{(s)}\|_2 + \epsilon} \quad (13)$$

The hypercolumn descriptor is formed by concatenation and final L_2 normalization:

$$\mathbf{H}_{i,j} = \text{Norm}\left(\hat{\mathbf{f}}_{i,j}^{(2)} \oplus \hat{\mathbf{f}}_{i,j}^{(3)} \oplus \hat{\mathbf{f}}_{i,j}^{(4)}\right) \quad (14)$$

yielding a dense feature grid $\mathbf{H} \in \mathbb{R}^{D \times H_2 \times W_2}$.

For sub-pixel accuracy, we apply quadratic interpolation around each candidate match location using the second-order Taylor expansion of the local feature response, following the approach used in SIFT Lowe [2004]. Given that the final hypercolumn descriptors are normalized to lie on the unit hyper-sphere, the dot product can be used to directly compute the cosine similarity for matching between descriptors.

Initial matches are filtered using a model fitting strategy. RANSAC estimates a projective homography H^* that minimizes reprojection error:

$$\epsilon = \|\mathbf{p}_{\text{fix}} - \pi(H\mathbf{p}_{\text{mov}})\|_2^2 \quad (15)$$

retaining geometrically consistent inliers Lei et al. [2025b], Katrusha et al. [2025].

6 Methodology: Domain-Adaptive Self-Distillation

Direct application of DINO-style self-distillation to side-scan sonar (SSS) imagery is challenging due to acoustic-specific statistics (speckle noise, gain variations) and small-batch distributed training constraints. We present a physics-informed self-distillation framework adapted to these conditions.

6.1 Backbone and Initialization

We use ConvNeXt-Tiny Liu et al. [2022] as the encoder for its balance of capacity and efficiency. It modernizes CNNs with ViT-inspired elements (patchify stem, depthwise conv, inverted bottlenecks) while preserving translation invariance and locality, which suit texture-rich sonar data.

Pre-trained DINOv3 weights are unavailable for this architecture and ImageNet features risk negative transfer. We therefore train entirely from scratch on the underwater domain.

6.2 Physics-Guided Augmentations

Acoustic imagery is inherently corrupted by speckle noise, a granular interference pattern resulting from the coherent summation of backscattered waves Preciado-Grijalva et al. [2022]. To encourage the network to learn features invariant to this signal-dependent noise, we injected Gaussian noise into the input tiles. This acts as a regularization term, forcing the student network to recover coherent structural representations from corrupted inputs, conceptually similar to the denoising objectives explored in Preciado-Grijalva et al. [2022].

The intensity of a returned sonar signal $I(x, y)$ is governed by the physical interaction between the acoustic pulse and the seafloor, modeled as:

$$I(x, y) = \rho(x, y) \cdot \cos(\theta(x, y)) \cdot L(x, y) \cdot G_{sys}$$

where G_{sys} represents the variable system gain and receiver sensitivity Lei et al. [2025a]. To simulate variations in G_{sys} and the dynamic range of the receiver—common artifacts arising from operator adjustments or automatic gain control (AGC) during missions Burguera and Oliver [2016], we applied random Color Jitter. By randomly perturbing the brightness (gain) and contrast (dynamic range) of the spectrograms, we enforce invariance to radiometric calibration errors, ensuring that the embedding space relies on the intrinsic seabed reflectivity ρ rather than transient intensity scaling Lei et al. [2025b].

6.3 Training Stabilisation and Loss Modifications

Training runs on two NVIDIA Quadro RTX 6000 GPUs with an effective batch size of 170 (85 per GPU $\times$ 2 views).

The original DINOv3 high-dimensional output ($K = 65,536$) caused near-zero gradients under sparse batches. We reduced the projection head output to $K = 128$ to ensure meaningful prototype occupancy and stable gradients.

The Sinkhorn-Knopp algorithm in the DINO loss frequently produced zero rows/columns, leading to NaNs. We replaced it with the DINOv1 centering mechanism for the global-to-local distillation loss Caron et al. [2021]:

$$C_t \leftarrow mC_{t-1} + (1 - m) \frac{1}{B} \sum_{i=1}^B O_{\text{teacher}}(x_i) \quad (16)$$

The teacher output center C is subtracted before softmax. This prevents collapse without Sinkhorn’s strict constraints. The iBOT masked image modeling head retains Sinkhorn, as patch tokens provide sufficient density.

6.4 Preliminary Qualitative Results

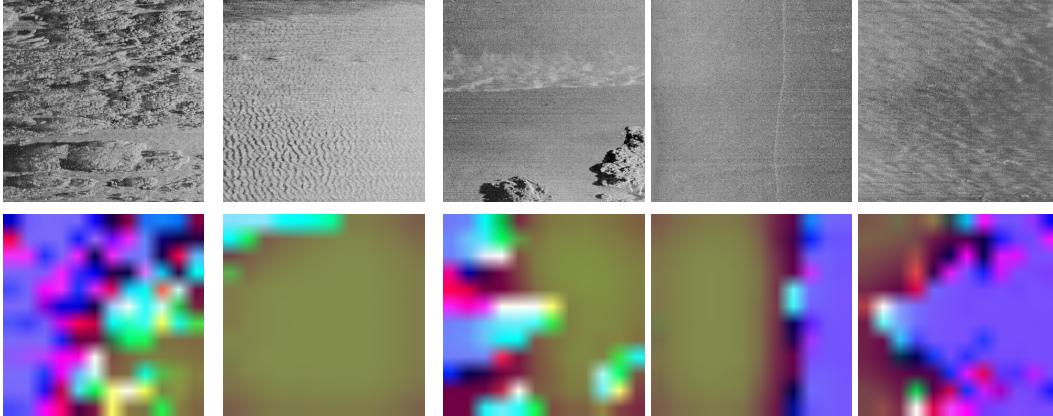


Figure 1: PCA Visualizations of the feature vectors extracted from the trained ConvNeXt backbone.

Due to the significant computational overhead required for the computation of nearest-neighbor benchmarks and the strict time constraints of the study, a full quantitative evaluation (e.g., k-NN probing or linear evaluation) was not feasible at this stage. However, qualitative inspection of the feature activation maps confirmed that the model began to successfully encode high-level semantic features. The network demonstrated an ability to consistently cluster consistent seafloor textures—distinguishing between sand ripples, rock outcrops, and acoustic shadows—despite the heavy application of noise and jitter augmentations. This indicates that the domain-adaptive self-distillation process successfully initiated the learning of physics-consistent representations from the *BenthiCat* dataset Rajani et al. [2025], overcoming the limitations of the training hardware.

7 Conclusion and Future Work

Side-scan sonar remains the primary modality for high-resolution seafloor mapping in subsea exploration Burguera and Oliver [2016], Rajani et al. [2025]. However, traditional approaches are limited by viewpoint-dependent backscatter, multiplicative speckle noise Lei et al. [2025b], Katrusha et al. [2025], and the severe scarcity of labeled data due to the high cost and complexity of marine acquisition Rajani et al. [2025], Khan et al. [2025], Preciado-Grijalva et al. [2022]. Self-supervised learning addresses these constraints by learning semantically and geometrically meaningful features directly from unlabeled raw data Siméoni et al. [2025], Khan et al. [2025].

7.1 Future Directions

Several directions remain for improving the proposed framework:

- Replace the Lambertian assumption with more advanced scattering models (e.g., Kirchhoff approximation or roughness-dependent empirical models) to better handle specular reflections and frequency-dependent effects from complex seafloor types Lei et al. [2025b,a].
- Incorporate physics-informed losses based on side-scan sonar physical models to further inform self-supervised representations.
- Optimize high-capacity models for edge deployment on AUVs through quantization and real-time inference techniques Liu et al. [2022].
- Develop more robust non-rigid registration methods to handle irregular overlap and cumulative navigation drift in real missions Lei et al. [2025b,a].

Advancing these areas will enable fully autonomous real-time seafloor classification and high-precision marine monitoring at scale Rajani et al. [2025].

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